REPORT NO. 03: USER STORY (CHAPTER-03)

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Abstract

This section of the Software Requirements Specification (SRS) defines the functional requirements and user stories for the proposed Multi-Tenant E-Commerce System. The purpose of this section is to clearly describe the core system functionalities, user interactions, and expected system behaviors that support online storefront management and wholesale product sourcing. The requirements focus on enabling small businesses and social-media-based sellers to create and manage digital stores, handle product listings, process orders, manage customers, and access supplier marketplaces within a shared cloud-based environment.

Through structured functional requirements and user stories, this section outlines how different user roles—such as store owners, customers, and administrators—will interact with the system. It specifies key features including tenant registration, product and inventory management, order processing, payment handling, analytics, and system administration. The goal is to ensure that the system meets user needs, improves operational efficiency, and supports scalable growth across multiple tenants while maintaining usability, reliability, and secure access.

1. OBJECTIVE

The main objectives of this document are as follows:

1. To establish a clear and structured Software Requirements Specification (SRS) for the proposed multi-tenant e-commerce system.
2. To identify and define the intended users, stakeholders, and primary use cases of the platform.
3. To analyze the overall product scope, operating environment, assumptions, and system constraints during the early stages of design.
4. To document the functional requirements, user interactions, and system behaviors without focusing on implementation or technical design details.
5. To ensure alignment between business objectives and system requirements through organized and comprehensive requirement analysis.

2. EQUIPMENT USED

The following tools and resources were utilized to prepare and complete this Software Requirements Specification (SRS) document:

1. A personal computer or laptop with internet access.
2. Web browsers (e.g., Google Chrome, Mozilla Firefox, Brave) for research and collaboration.
3. Word processing and text editing software (e.g., Notepad, text editors, MS Word, or LaTeX) for drafting and formatting the SRS.
4. Software Requirements Specification (SRS) guidelines and templates provided through Google Classroom.
5. Discord server for team communication, discussion, and coordination.
6. Trello for task management, progress tracking, and project organization.
7. Course materials related to Object-Oriented Analysis and Design (OOAD) for reference and requirement analysis.

3. Functional Requirements and User Stories

The following chapter outlines the core functional requirements, user stories, and feature assignments for the Multi-Tenant E-Commerce System. Each requirement is linked to a responsible team member for implementation.

3.1 Functional Modules and Responsibilities

1. **Sign-Up and Security Management (Shihab)**
   - **Features:**
     - Two-Factor Authentication (2FA) during registration and login.
     - Vendor certificate verification and Tax Identification Number (TIN) validation.
     - Shop deletion or deactivation functionality.
     - Secure storage of sensitive information in the database.
   - **User Stories:**
     - As a vendor, I want to securely register with 2FA so that my account is protected.
     - As a Super Admin, I want to verify vendor certificates and TINs to ensure only legitimate businesses are onboarded.
     - As a vendor, I want the option to delete my shop if I stop operations, ensuring control over my data.
2. **Offer and Theme Customization (Rohan)**
   - **Features:**
     - Creation and management of promotional offers and discounts.
     - Customization of storefront themes, including colors, logos, and banners.

- **User Stories:**
  - As a vendor, I want to design my store with custom themes to maintain brand identity.
  - As a vendor, I want to run time-limited offers so that I can attract more customers.

3. **Payment Gateway Integration and Product Tracking (Lineya)**
   - **Features:**
     - Integration with predefined payment gateways: bKash, Nagad, Cash on Delivery (COD), Visa, MasterCard.
     - Tracking product orders, including status updates from processing to delivery.
   - **User Stories:**
     - As a customer, I want multiple secure payment options so that I can pay conveniently.
     - As a vendor, I want to track the status of each order to manage deliveries efficiently.
4. **Inventory Management and Analytical Dashboard (Marufa)**
   - **Features:**
     - Maintain and update product inventory in real-time.
     - Analytical dashboard to monitor sales, revenue, and inventory trends.
   - **User Stories:**
     - As a vendor, I want my inventory automatically updated with each order to prevent overselling.
     - As a vendor, I want a dashboard displaying sales trends to make informed business decisions.
5. **Promotional Notifications and Feedback Management (Taskif)**
   - **Features:**
     - Sending promotional notifications to customers for offers or product updates.
     - Collecting and managing customer feedback, including ratings and reviews.
   - **User Stories:**

  - As a vendor, I want to send notifications to my customers to increase engagement and sales.
  - As a vendor, I want to view and respond to customer reviews to improve service quality.

This chapter establishes the core functional areas of the Multi-Tenant E-Commerce System, aligning each requirement with specific user stories and responsible team members. This structured assignment ensures clarity in development, accountability, and comprehensive coverage of the system's functional expectations.

4. My Contributions (Functional Modules: Inventory Management and Analytical Dashboard)

In this project, I was responsible for the design, analysis, and documentation of two critical functional modules for the Multi-Tenant E-Commerce System:

1. **Inventory Management**
2. **Analytical Dashboard (Analytics Board)**

These modules are essential for enabling vendors to efficiently manage their online storefronts, monitor stock levels, and make informed business decisions using data-driven insights.

4.1 Inventory Management (IM-07)

Description: The Inventory Management module provides vendors with comprehensive control over their product lifecycle and stock accuracy. It ensures that the digital storefront remains synchronized with physical inventory, preventing overselling and enhancing operational efficiency.

Title: Inventory Management

Description: This feature provides vendors with full control over their product lifecycle and stock accuracy, ensuring the digital storefront remains synchronized with physical availability.

Specific Requirements:

ID	Priority	Requirement
IM-07.1	High	The system shall allow vendors to input and store product names, prices, SKUs, and quantities in a central database.
IM-07.2	High	The system shall automatically decrement stock levels upon order placement and update the vendor dashboard in real-time.
IM-07.3	Medium	The system shall provide an interface for editing or removing products, ensuring the storefront reflects changes instantly.
IM-07.4	Medium	The system shall implement validation rules to prevent stock quantities from falling below zero during manual updates.
IM-07.5	Low	The system shall allow vendors to set thresholds and trigger automated dashboard notifications when stock falls below the limit.

Figure 4.1: Inventory Management table

Key Features and Requirements:

- Vendors can input and store product names, SKUs, prices, and quantities in a centralized database (IM-07.1).
- Stock levels are automatically decremented when customers place orders, updating the vendor dashboard in real-time (IM-07.2).
- Vendors can edit or remove products, with instant reflection on the storefront (IM-07.3).
- Validation rules prevent stock from falling below zero during manual updates (IM-07.4).
- Vendors can set stock thresholds and receive automated notifications when items reach low stock levels (IM-07.5).

User Story:

US-07.1: Product and Stock Lifecycle Management
As a Vendor, I want to add, update, and monitor my product stock levels

with real-time automated syncing, so that I can prevent overselling and maintain an accurate online store.

User Story for IM-07:

US-07.1: Product and Stock Lifecycle Management

As a Vendor,

I want to add, update, and monitor my product stock levels with real-time automated syncing,

So that I can prevent overselling and maintain an accurate online store.

Acceptance Criteria:

- Products are added with mandatory SKU and quantity fields.
- Stock levels decrease immediately after a successful customer checkout.
- Vendors receive a dashboard alert when items reach the defined "Low Stock" threshold.

Figure 4.2: Inventory Management (User Story)

Acceptance Criteria:

- Products are added with mandatory SKU and quantity fields.
- Stock levels decrease immediately after a successful customer checkout.
- Vendors receive a dashboard alert when items reach the defined ”Low Stock” threshold.

Professional Significance: The Inventory Management module is crucial for ensuring smooth business operations and avoiding stock-related issues. It helps vendors maintain credibility with customers by preventing order failures due to stock discrepancies. This module also reduces manual effort and provides automated alerts, allowing vendors to focus on growth strategies rather than routine stock tracking.

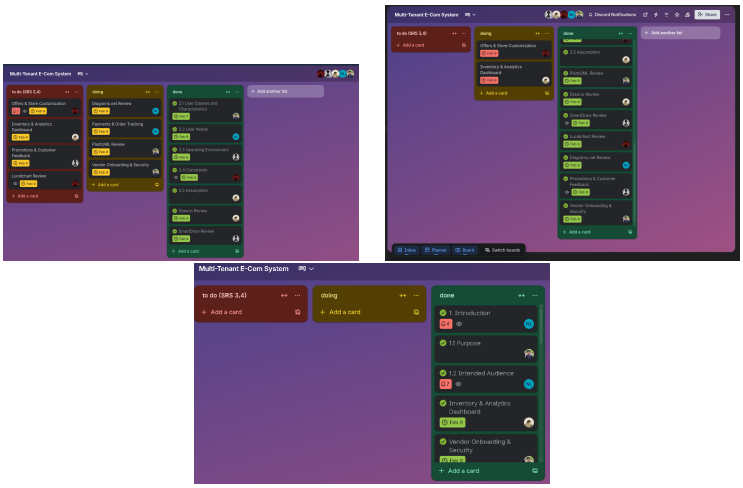

Figure 4.3: Inventory Management Module: (a) Product list and stock entry interface, (b) real-time stock update after order placement, (c) low-stock alert and inventory monitoring dashboard.

4.2 Analytical Dashboard (AB-08)

Description: The Analytical Dashboard module translates raw sales data into actionable business insights through interactive visualizations. Vendors can monitor sales performance, revenue trends, and product popularity to make informed decisions that drive revenue growth.

Title: Analytical Board (Analytics Dashboard)

Description: This feature translates raw sales data into actionable business insights through visualization, helping vendors make data-driven decisions.

Specific Requirements:

ID	Priority	Requirement
AB-08.1	High	The system shall calculate and display total sales, revenue, and order counts updated in real-time on the dashboard.
AB-08.2	Medium	The system shall generate a ranked list of best-selling items, including specific sales counts and revenue generated per product.
AB-08.3	Medium	The system shall render sales data using bar charts (for revenue trends) and pie charts (for product distribution).
AB-08.4	Low	The system shall provide a date-picker utility to filter all dashboard visualizations and metrics by a user-defined timeframe.

Figure 4.4: Analytical Dashboard table

Key Features and Requirements:

- Display total sales, revenue, and order counts in real-time (AB-08.1).
- Generate a ranked list of best-selling products with sales counts and revenue per item (AB-08.2).
- Render data using bar charts (revenue trends) and pie charts (product distribution) (AB-08.3).
- Provide a date-picker utility to filter all visualizations by a user-defined timeframe (AB-08.4).

User Story:

US-08.1: Sales Data Visualization and Insights
As a Vendor, I want to view visual charts and filtered reports of my sales performance, so that I can identify trends and determine which products are driving my revenue.

User Story for AB-08:

US-08.1: Sales Data Visualization and Insights

As a Vendor,

I want to view visual charts and filtered reports of my sales performance,

So that I can identify trends and determine which products are driving my revenue.

Acceptance Criteria:

- Dashboard displays real-time key performance indicators (KPIs).
- Visual charts (Bar/Pie) accurately represent the data stored in the database.
- Filtering by date updates all charts and lists without requiring a full page reload.

Figure 4.5: Inventory Management(User Story)

Acceptance Criteria:

- Dashboard displays real-time key performance indicators (KPIs).
- Visual charts (Bar/Pie) accurately represent the stored sales data.
- Filtering by date updates all charts and lists dynamically without requiring a full page reload.

Professional Significance: This module empowers vendors to make strategic business decisions by providing insights into sales trends, high-performing products, and revenue generation. It helps identify which products are popular and which need promotional support. By visualizing sales data, the dashboard reduces reliance on manual calculations, increases operational efficiency, and provides a clear understanding of business health.

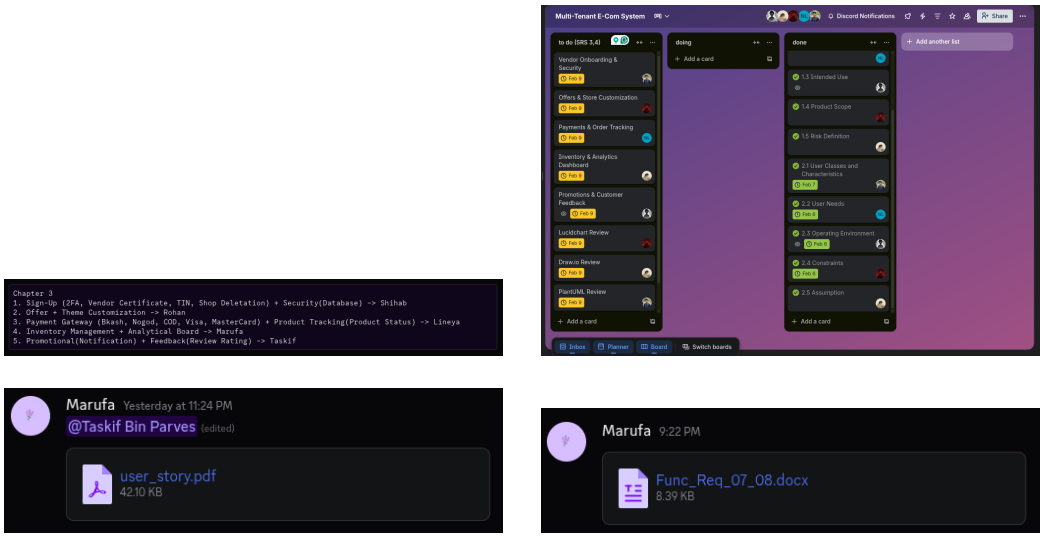

Figure 4.6: Analytical Dashboard Module: (a) real-time KPI overview, (b) revenue trend bar chart, (c) product distribution pie chart, (d) filtered sales report by date range.

Together, the Inventory Management and Analytical Dashboard modules form the backbone of the vendor's operational workflow. The inventory module ensures accurate product availability, while the analytics module provides data-driven insights that enhance decision-making and profitability. Both modules are essential for professional e-commerce management and reflect my individual contribution to the project.

5. DISCUSSION AND PROOFS

This chapter presents discussion and visual proof of the activities performed during the preparation of the Software Requirements Specification (SRS). Trello was used for task management and progress tracking, while Discord was used for communication, discussion, and file sharing.

These tools helped maintain organization, ensured timely completion of tasks, and supported proper coordination during the preparation of Section 1.5 (Risk Definition) and Section 2.5 (Assumptions).

5.1 Trello Activity Proof

Trello was used to manage and track tasks related to the SRS documentation. Tasks were organized into **To Do**, **Doing**, and **Done** categories to monitor progress and ensure accountability among team members.

The screenshots below demonstrate task assignment, progress tracking, and completion status throughout the project.

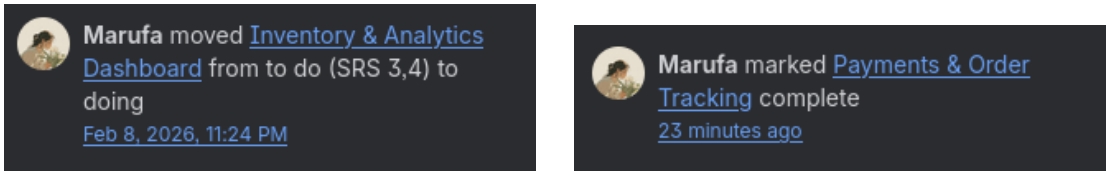

Figure 5.1: Trello board showing task planning, progress tracking, and completed activities

5.2 Discord Communication Proof

Discord was used as the primary communication platform for discussion, coordination, and file sharing among team members. It allowed real-time updates, clarification of

tasks, and quick decision-making during the preparation of the SRS document.

The screenshots below show communication related to task assignments, discussions, and submission coordination.

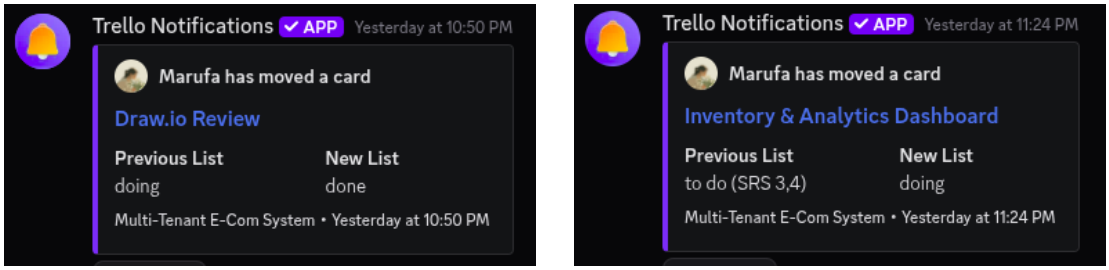

Figure 5.2: Discord communication showing discussion, coordination, and file sharing among team members

6. CONCLUSIONS

This chapter summarized the functional requirements and user stories defined in Chapter 3 of the Software Requirements Specification (SRS) for the proposed Multi-Tenant E-Commerce System. The requirement analysis process identified the core system functionalities necessary to support vendors, customers, and administrators within a shared cloud-based platform.

Through structured functional requirements, the system's expected behaviors—such as secure sign-up, inventory management, payment processing, analytics dashboards, notifications, and customization—were clearly defined. Each feature was translated into user stories to ensure that the system design remains user-centered and aligned with real-world business workflows. These user stories help bridge the gap between technical development and user expectations by describing how different stakeholders will interact with the system to accomplish their goals.

The documentation of requirements also ensured traceability, clarity, and accountability by assigning specific functional modules to team members. This structured approach supports systematic development, simplifies testing and validation, and reduces misunderstandings during implementation.

Overall, Chapter 3 provides a comprehensive foundation for the system's development by defining what the system must do and how users will interact with it. The clearly documented requirements and user stories will guide the design, development, and testing phases, ensuring that the final system is scalable, secure, and responsive to the needs of small and social-commerce-based businesses.